Secondary Alveolar Bone Grafting and Iliac Cancellous Bone Harvesting for Patients With Alveolar Cleft

Weiyi Pan, MD, DDS, Chenzhou Wu, MD, DDS,* Zheng Yang, BE,† Zexi Duan, MD, DDS,* Zhifei Su, MD, DDS,* Peiqi Wang, MD, DDS,* Qian Zheng, DDS, PhD,‡ and Chunjie Li, DDS, PhD§*

Abstract: To assess the efficacy of present interventions optimizing the result of secondary alveolar bone grafting (SABG) and the interventions alleviating the donor site morbidity after iliac cancellous bone harvesting. Researches were identified by searching the electronic database of MEDLINE, EMBASE, Cochrane Central Register of Controlled Trials, Chinese BioMedical Literature Database, and the China National Knowledge Infrastructure. In addition, relevant journals and references of the included studies were searched manually. The Oxford 2011 Levels of Evidence were applied to assess the methodological quality of selected studies, and the best evidence synthesis system was applied afterward to measure the strength of evidence. As a result, 42 studies were considered eligible and included, among which 4 were of high quality while 38 were of low quality. Thirty lines of evidences were acquired after the synthesis, among which 13 were rated as moderate while 17 were rated as insufficient. As for the interventions optimizing the result of SABG, moderate evidence confirmed the efficacy of preoperative orthodontic treatment, the superiority of performing SABG before the eruption of canine, and the accuracy of cone beam computed tomography in preoperative estimation of the cleft volume. As for the interventions alleviating the morbidity of iliac cancellous bone harvesting, moderate evidence confirmed the treatment benefit of the interventions below: minimally invasive technique, including trephine and Shepard osteotomy; preemptive analgesia, including continuous bupivacaine infusion or transversus abdominis plane block. As for the rest interventions, only insufficient evidence was found.

Key Words: Alveolar cleft, best evidence synthesis, iliac bone harvesting, secondary bone grafting, systematic review

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From the *State Key Laboratory of Oral Diseases, West China Hospital of Stomatology, Sichuan University; †School of Information and Software Engineering, University of Electronic Science and Technology of China; ‡Department of Cleft Lip and Palate; and §Department of Oral and Maxillofacial Surgery, State Key Laboratory of Oral Diseases, West China Hospital of Stomatology, Sichuan University, Chengdu, China.
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Address correspondence and reprint requests to Chunjie Li, MD, DDS, PhD, Department of Oral and Maxillofacial Surgery, State Key Laboratory of Oral Diseases, West China Hospital of Stomatology, Sichuan University, 610041 Chengdu, China;
E-mail: lichunjie07@qq.com
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Cleft lip or cleft lip and palate is usually accompanied by alveolar cleft,[1] with an incidence of 0.18 to 2.50 per 1000 births varying among races.[2] Alveolar cleft results from the incomplete infusion of intermaxillary prominence which can be caused by genetic and environmental factors. If not treated, it may result in oronasal fistula, voice disorder, maxillary growth disruption, crowded dentition, etc.[3] Reconstruction of alveolar cleft can reestablish the dental arch continuity, stabilize the premaxilla, provide bony support for the tooth eruption, and close the oronasal fistula, etc.[4–6]

In 1972, Boyne and Sands[7] proposed that performing secondary alveolar bone grafting (SABG) with iliac cancellous bone in the mixed dentition, namely from 7 to 11-year old, which could repair the alveolar cleft effectively without significant impact on maxillary growth. And the method has been widely accepted. Iliac cancellous bone harvested from iliac anterior crest is the most common grafting material for SABG. Even though some new methods were put forward, such as the application of recombinant human bone morphogenetic protein-2(rhBMP-2)[8,9] and other bone substitute materials,[10,11] iliac cancellous bone was still applied in most patients because of its bony abundance and ability to harvest simultaneously with recipient site preparation.[12] However, even though SABG has been acknowledged widely as an important part of treatment for cleft lip and palate, uncertainty and controversy still exist. Standards are still lacking concerning the design of recipient-site surgical flap and postoperative assessment. Preoperative orthodontic expansion and simultaneous osteotomy have been put forward to improve the alveolar reconstruction, while the clinical effect is uncertain. Furthermore, morbidities of iliac cancellous bone harvesting (ICBH), such as postoperative pain, abnormal gait, and hematoma formation, are severe,[13] which urges clinicians to find solutions to reduce the incident rate of them.

In addition, new therapeutic options in the field of alveolar cleft reconstructions have been put forward in recent years. New design of surgical flaps was proposed,[14] and new radiological imaging technique was applied to evaluate the success rate of SABG.[15] Besides, clinicians began applying nontransitional surgical approaches[16] and preemptive analgesia[17] to alleviate the morbidity of ICBH.

The aim of present systematic review is to assess the efficacy of present interventions to optimize the result of SABG and reduce the morbidity rate of ICBH.

METHODS

The study inclusion, quality assessment, data extraction, and best evidence synthesis (BES) were performed independently by 2 researchers (WP, CW), and disagreements were resolved

TABLE 1. Inclusion Criteria of Interventions and Outcomes

Interventions	Primary Outcomes
Recipient-site interventions	
Preoperative orthodontics	Alveolar reconstruction*
Timing of surgery	Alveolar reconstruction
Recipient-site flap design	Alveolar reconstruction
	Complication rate†
Simultaneous osteotomy	Alveolar reconstruction
	Maxillary growth
	Complication rate
Wound management	Alveolar reconstruction
	Complication rate
Radiographic imaging assessment	Efficacy of grafts assessment
Donor-site interventions	
Minimally invasive	Operability of the technique‡
Surgical approaches	Complication rate
Donor-site analgesia	Morbidity§
Outpatient surgery	

*Alveolar reconstruction includes outcomes as follows: bone height or clinical success rate, bone density, bone resorption/formation rate, etc.

†Complications include conditions as follows: postoperative bleeding, infection, wound dehiscence, urinary retention, hypertrophic scarring, delayed healing, etc.

‡Operability of the technique includes outcomes as follows: operating time, intraoperative blood loss, etc.

§Morbidity includes outcomes as follows: incidence of morbidity, efficiency of pain relief (including the pain scores and postoperative analgesic usage), time to initial ambulation, length of stay in hospital, etc.

through discussion. The protocol was designed and registered in PROSPERO (CRD42014009942) before the present systematic review.

Inclusion Criteria

The inclusion criteria were as follows. First, the included studies were required to be systematic reviews or meta-analysis, randomized controlled trails, clinical controlled trials, and cohort studies. Second, all the participants had to be alveolar cleft patients who planned to undergo or had undergone SABG. Third, the control group had to undergo standard SABG and traditional anterior ICBH, while no limitation was set in intervention groups. Finally, the primary outcome was defined according to the interventions, as listed in Table 1. In addition, studies investigating the cost of specific interventions would be included as well.

Search Strategy

The chosen electronic databases, including MEDLINE (via OVID, 1948 to May 2015), EMBASE (via OVID, 1984 to May 2015), Cochrane Central Register of Controlled Trials (CENTRAL, issue 5, 2015), Chinese Bio Medical Literature Database (CBM, 1978 to May 2015), and the China National Knowledge Infrastructure (CNKI, 1994 to May 2015), were searched for published, unpublished, and ongoing studies. No language or time limitations were set in the database searching. Moreover, relevant journals and references of included studies were searched manually.

The search strategy combined MeSH with free text words. The MeSH terms used were as follows: "CLEFT PALATE," "CLEFT LIP," "BONE TRANSPLANTATION," and "ILIUM." The possible eligible studies were determined preliminarily by scanning the titles and abstracts of all studies resulting from the search. The full texts of the possible eligible studies were obtained for final judgment.

Quality Assessment

The methodological quality of included studies was assessed by using the "treatment benefit" part of "The Oxford 2011 Levels of Evidence" from Oxford Centre for Evidence-Based Medicine.[18] Studies graded Levels 1 and 2 were defined as "high quality," and studies graded Level 3 and below were defined as "low quality."

Data Extraction

Information as below was extracted: number of patients, including the sex ratio; mean age (the overall mean age of the studies investigating the timing of surgery was considered pointless and not extracted); type of cleft, including the proportion of bilateral patients; interventions and controls, including the number of patients in each group; primary outcomes, including the *P* value.

Level of Scientific Evidence

To synthesize the methodological quality of the studies and to be able to draw conclusions on the effects of alternative interventions, we applied Best Evidence Synthesis (BES).[19,20] This rating system takes into account the number of studies, methodological quality, and consistency of outcomes of the studies. The rates are as follows:

1. Strong evidence, provided by generally consistent findings in multiple ($\geq$2) high-quality studies;
2. Moderate evidence, provided by generally consistent findings in 1 high-quality study and 1 or more low-quality studies, or in multiple low-quality studies;
3. Insufficient evidence, when only 1 study was available or findings were inconsistent in multiple ($\geq$2) studies.

We considered results to be consistent when at least 75% of the studies showed results in the same direction, which was defined according to significance ($P < 0.05$). If the *P* value was not given, it will be noted specially. If 2 or more studies were of high methodological quality, we disregarded the studies of low methodological quality in the BES.

RESULT

Search and Study Inclusion

One thousand three hundred thirteen publications were identified through electronic-database searching and manual searching, of which 1256 were excluded after screening the title and abstracts. The full texts of the remaining 63 studies were obtained, while only 42 of them were considered eligible. The process of study identification is presented in Figure 1.

Characteristics and Methodological Quality of Included Studies

The characteristics and methodological qualities are listed in Supplemental Digital Content, Table E1, http://links.lww.com/SCS/A196. The studies aiming at recipient-site or donor-site interventions are separated.

The 42 included studies[21–62] consisted of 4 high-quality studies[39,54,56,58] and 38 low-quality studies. Among which, 1 randomized controlled trail[39] was downgraded since the sequence generation, allocation concealment, and blinding were all not clearly reported. In addition, 2 studies[53,62] included patients or applied interventions that did not meet our inclusion criteria. Not all arms of the studies met our inclusion criteria. However, since the outcome was reported separately, the arms that meet our inclusion criteria were included without downgrading (see Supplemental Digital Content, Tables E1 and E2, http://links.lww.com/SCS/A196 and http://links.lww.com/SCS/A197).

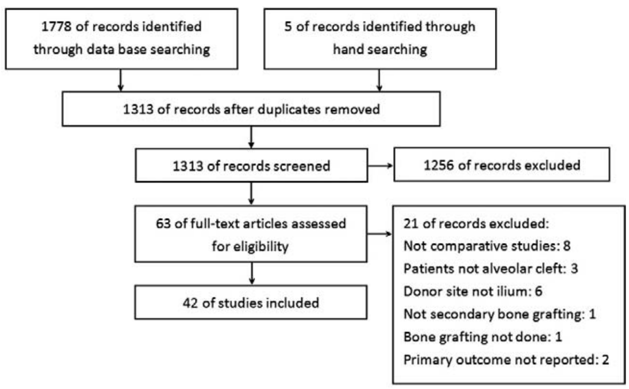

FIGURE 1. Flow of study inclusion.

Best Evidence Synthesis

The results of BES are listed in Tables 2 and 3.

Thirty lines of evidences were acquired after the BES, among which 13 were rated as moderate while 17 were rated as insufficient. One low-quality study compared 2 methods of minimally invasive technique with traditional ICBH, so the BES was performed separately, since the results were reported separately.

Among the interventions aiming to optimize the result of SABG, only the treatment benefit of preoperative orthodontic treatment was supported by moderate evidence. Besides, performing SABG before the eruption of canine showed certain superiority in all types of cleft, which was supported by several lines of moderate evidence. In addition, CBCT was found effective in preoperative estimation of the cleft volume, which was supported by moderate evidence as well. As for the rest interventions, only insufficient evidence was found.

Among the interventions aiming at alleviating the morbidity of ICBH, moderate evidence confirmed the treatment benefit of the interventions below: minimally invasive technique, including trephine and Shepard osteotomy; preemptive analgesia, including continuous bupivacaine infusion or transversus abdominis plane block. As for the rest interventions, that is, minimally invasive technique and preemptive analgesia, as well as iliac postoperative approach and ambulatory operation, only insufficient evidence was found.

DISCUSSION

Interventions for Recipient Site

Secondary alveolar bone grafting had been acknowledged widely since it was proposed in 1972. However, there has been a long-term debate about when SABG should be performed, and how the recipient site should be prepared. Besides, the efficacy and the impact on alveolar reconstruction and maxillary growth of preoperative orthodontic and orthognathic treatment are uncertain. Moreover, no standard has been established in postoperative assessment. In this systematic review, we summarized all related clinical evidence to find these answers.

Preoperative Orthodontics

Lingual tipping and rotation of the front teeth may occur in patients with cleft lip and palate, especially bilateral cleft lip and palate (BCLP) patients, and the bone segments of alveolar process are often too close to perform SABG.[63] In 1964, Matthews and Grossman[64] reported that preoperative orthodontic treatment could create good conditions for the operation, which has already been affirmed to be effective for alveolar reconstruction. At present, preoperative orthodontics has already become a normal procedure of SABG treatment.

In this systematic review, we found moderate evidence that the success rate of preoperative orthodontics group is higher than nonorthodontics group.[21,22] However, it is worth attention that nonorthodontics group involved patients who did not receive preoperative orthodontics because of poor dental health and those who did not require preoperative orthodontics, which may cause the deviation in result.[21]

Timing of Surgery

Primary bone grafting can repair the alveolar cleft and close oronasal fistula when patients are young, while its impact on maxillary growth limits the extensive clinical adoption.[65,66] Ever since performing SABG in mixed dentition was proposed in 1972, studies have confirmed its little impact on maxillary growth.[67] At present, it has been widely recognized and applied clinically. Since the alveolar cleft occurs mainly in the bone segments of lateral incisor or canine, the eruption of the teeth, especially canine, is one of the most important factors of choosing the timing of SABG. To be specific, the eruption of canine in cleft area may disturb mucosal dissection and bone grafting, which may lead to poorer success rate.[24,25]

After BES, we found moderate evidence that bone grafting before the eruption of canine can improve the success rate of unilateral cleft lip and palate and BCLP patients, while the effect on cleft lip and alveolar process patient is not significant.[23–29] The result may be caused by the relatively more soft tissue for covering the grafted bone in cleft lip and alveolar process patients than complete cleft lip and palate patients.[25] Moderate evidence also supports that performing SABG during mixed dentition or at younger age could result in better alveolar reconstruction.[30–33]

Design of Surgical Flap and Postoperative Management of Surgical Wound

The operation technique of recipient site is extremely important, and is considered the key point to determine the success of SABG.[68,69] As for the component of tissue flap, mucosal flap and mucoperiosetal flaps are commonly applied. Mucoperiosteal flap is more favorable than mucosal flap, as the presence of periosteum can improve the ossification and survival rate of grafts. In the present systematic review, we found insufficient evidence that using mucoperiosteal flaps shows higher clinical success rate and lower postoperative complications incidence rate than mucosal flap.[34]

As for the incision design, there are 2 common approaches: the labial and palatal double-incision approach described by Turvey et al;[70] the labial single-incision approach described by Abyholm et al.[71] Both of them need to remove the excessive soft tissue in the cleft. Removing the extra soft tissue may cause the deficiency of soft tissue for tension-free closure at the labial side, which leads to the requirement of relaxant incision and tension-release suture consequently.[72,73] Palate side approach has been promoted recently.[74] The incisions are made at the palatal side of the cleft, and the tissues in the cleft are made full use of to insure the closure of nasal floor and labial side, so as to achieve the tension-free closure of labial incision without requirement of relaxant incision. Besides, it improves the bone fill rate at the palatal side. However, we failed to find studies comparing the efficacy of the approaches mentioned above.

Studies on operation methods of bone grafting are mainly patient reports, and fail to meet the inclusion criteria of this systematic review. The main reason is that the clinicians select specific surgical

TABLE 2. Results of Best Evidence Synthesis (Recipient-Site Interventions)

Level of Scientific Evidence	Synthesis of Evidence	Investigator (Methodological Quality)
Preoperative orthodontic expansion		
Moderate	Performing preoperative orthodontic expansion could improve the alveolar reconstruction	Kindelan[21] (low)
		McIntyre[22] (low)
Timing of surgery		
Moderate	For UCLP and BCLP patients, performing SABG before eruption of canines could result in better alveolar reconstruction	Enemark[23] (low)
	For CLAP patients, timing of surgery showed no influence on the alveolar reconstruction	Jia[24] (low)
		Sindet-Pedersen[25] (low)
	Performing SABG before eruption of canines could result in better alveolar reconstruction	Helms[26] (low)
		Rawashdeh[27] (low)
	For bilateral patients, performing SABG before eruption of canines could result in better alveolar reconstruction	Mao[28] (low)
		Jia[29] (low)
	Performing SABG during mixed dentition could result in better alveolar reconstruction	Terheyden[30] (low)
		Dempf[31] (low)
	Performing SABG at younger age could result in better alveolar reconstruction	Lu[32] (low)
		He[33] (low)
Flap design for closure of alveolar cleft		
Insufficient	Using Mucoperiostal flap could result in less postoperative dehiscence, less need of secondary periodontal surgery and more fistula relapse, with no influence on eruption of canine	Hugentobler[34] (low)
Simultaneous osteotomy		
Moderate	Simultaneous osteotomy would affect alveolar reconstruction	Freihofer[35] (low)
		Samman[36] (low)
Insufficient	For BCLP patients, simultaneous osteotomy is conductive to alveolar reconstruction and maxillary growth	Akita[37] (low)
		Jensen[38] (low)
Recipient-site postoperative additional procedures		
Insufficient	Using continuous local cooling could reduce the incidence of complications and alleviate the recipient-site postoperative pain	Wang[39] (high)
Radiographic imaging evaluation		
Moderate	Using CBCT preoperatively to estimate the cleft volume is reliable	Hu[40] (low)
		Xu[41] (low)
Insufficient	Anterior occlusal radiograph and intraoral panograph are similar in evaluation of bone height	Jia[42] (low)
	Using intraoral films could result in overestimation of bone graft status	Iino[43] (low)
	Using 3D-CT preoperatively to estimate the cleft volume is reliable	Xi[44] (low)

BCLP, bilateral cleft lip and palate; CBCT, cone beam CT; CLAP, cleft lip and alveolar process; CT, computed tomography; SABG, secondary alveolar bone grafting; UCLP, unilateral cleft lip and palate.

methods according to the individual conditions of patients during clinical practices because of the indications the surgical flaps have.

As for the interventions alleviating the recipient-site postoperative pain, only 1 study applying continuous local cooling was included. The biological effects caused by local cooling include local analgesia, inhibited edema formation, and reduced blood circulation.[75] We found insufficient evidence that using continuous local cooling could reduce the incidence of complications and alleviate the recipient postoperative pain. It is worthy of further research since it is simple to operate, high performing, and cost-effective.

Simultaneously Osteotomy

It has been reported that maxillary hypoplasia may occur after the primary cleft lip and palate repair, which appear as midfacial growth deficiency, depression of nasal base area, and following orthognathic operation is required.[76,77] To solve this problem, Brouns and Egyedi[78] proposed performing osteotomy simultaneously with SABG to reduce the number of operations and alleviate the financial burden of patients.[35,36,79]

In the included studies, there are completely opposite opinions on whether the osteotomy would impact the success rate of the bone

TABLE 3. Result of Best Evidence Synthesis (Donor-Site Interventions)

Level of Scientific Evidence	Synthesis of Evidence	Investigator (Methodological Quality)
Surgical approaches to iliac bone harvesting		
Insufficient	Using posterior approaches to harvest iliac bone could reduce the blood loss and length of stay in hospital, while duration of the surgery is longer, and it would not cost more	Abramowicz[45] (low)
		Kupfer[46] (low)
Minimally invasive technique		
Moderate	Using trephine is simple and could alleviate morbidity, without extra risk of complications	Sharma[47] (low)
		McCanny[48] (low)
		Burstein[49] (low)
	Using Shepard osteotome could alleviate the morbidity	Eufinger[50] (low)
		Constantinides[51] (low)
Insufficient	Using percutaneous hollow needle is simple and could alleviate morbidity, without extra risk of complications	Hardy[52] (low)
	For UCLP patients, using percutaneous hollow needle could alleviate morbidity and reduce the complications	Gimbel[53] (low)
	Using trephine has no effect on clinical success rate	McCanny[48] (low)
	Using bone grinder is simple and could alleviate morbidity, without extra risk of complications	Burstein[49] (low)
	Using Shepard osteotome is simple, and would not affect clinical success rate and incidence of complications	Constantinides[51] (low)
Management of postoperative donor-site pain		
Moderate	Continuous bupivacaine infusion could alleviate morbidity	Meara[54] (high)
	Using transversus abdominis plane block could alleviate morbidity	Sbitany[55] (low)
		Mehra[56] (high)
		Tanaka[57] (low)
Insufficient	Using ketorolac intravenously could not reduce the analgesic usage.	Dawson[58] (high)
	Intraoperative bupivacaine infiltration could alleviate morbidity	Hoard[59] (low)
	Continuous bupivacaine infusion would not cause extra risk of complications	Sbitany[55] (low)
	Bupivacaine-soaked absorbable gelatin sponge could alleviate morbidity, without extra risk of complications.	Dashow[60] (low)
Ambulatory alveolar bone grafting		
Insufficient	Performing outpatient SABG would not lead to extra risk of complications, and the total cost is less	Perry[61] (low)
		Albert[62] (low)

SABG, secondary alveolar bone grafting; UCLP, unilateral cleft lip and palate.

grafting or not. We found moderate evidence that for patients with alveolar cleft, the simultaneous osteotomy could result in the declining success rate of bone grafting.[35,36] However, when only bilateral patients were involved, we found insufficient evidence that the simultaneous osteotomy could decrease the cleft size and increase the success rate.[37] As for the contradictions presented here, some scholars think that the decreasing success rate is mainly caused by the inappropriate operation technique which can be avoided.[80,81]

Besides, we find insufficient evidence that simultaneous osteotomy did not show impact on the maxillary growth.[38] It was confirmed by Padwa et al who proved that the early (6–8 year) osteotomy would not lead to the deficiency of midfacial growth. It implies that there is no need to wait for the complete growth of maxilla to perform the osteotomy.[82]

In addition, it is worth noting that simultaneous osteotomy is of special importance for patients failing to undergo SABG at proper time[36] because of significant maxillary growth deficiency. It is quite necessary to design studies to explore the clinical effect of simultaneous osteotomy on them.

Imaging Assessment

Effective bone mass (bone width, thickness, and height) is exactly the basic conditions of maintaining postoperative orthodontics teeth alignment and stabilization of the implant in the cleft area.[79] Bone height is the most important factor of bone mass, which serves as the standard of evaluating the success rate of SABG.[83,84]

X-ray plain film is widely applied clinically to evaluate the success rate of SABG. In this systematic review, only Jia et al[42] compared the anterior maxillary occlusal radiograph with intraoral panograph, and provided insufficient evidence that no essential distinctions were found in the estimate bone height.

As far back as 1987, clinicians have discovered through long-term follow-up that only 50% of patients who were defined as success by X-ray plain film achieved the real cleft closure.[23] The accuracy and reliability of the X-ray plain film have been questioned constantly. In addition, X-ray plain film has intrinsic disadvantages that cannot overcome, for instance, the failure in reflecting the volume information, or the overlapping and deformation of image.[85] Therefore, three-dimensional (3D) imaging techniques were applied to improve the accuracy of imaging assessment.

Computed tomography (CT) is the most widely used 3D imaging technique. Computed tomography has evident advantages in the evaluation of 3D configuration of bone bridge, especially the anteroposterior thickness of the bone bridge, which is important for orthodontic tooth alignment and the application of dental implants. However, only 1 study was included and provided insufficient evidence that when compared with CT, intraoral films could result in overestimation of bone graft status in part of patients.[43]

Cone beam CT (CBCT) is another 3D imaging technique that has been widely studied and applied in the fields of dentistry in recent years. Its imaging precision is about 8 times CT with only about 1/10 radiation. Besides, it is characterized by the short scanning time and low requirements of the position.[86,87] However, no studies comparing CBCT with other imaging techniques were included in this systematic review.

Along with the popularization of 3D imaging technique, preoperative evaluation of the alveolar cleft was promoted to predict the bone volume acquired for patients, especially BCLP patients, which contribute to the selection of optimal donor site and surgical methods. We found moderate evidence that CBCT is reliable in preoperative cleft volume estimation.[40,41] However, we only found insufficient evidence that 3D-CT is reliable in volume estimation, as only 1 included study reported 3D-CT could be used to predict cleft volume.[44]

Interventions for Donor Site

At present, ilium has already become the most widely applied donor site, for it can offer substantial cancellous bone that can be harvested easily. Besides, the operation on the recipient site can be performed simultaneously, which can shorten the duration of operation. However, ICBH is criticized for the postoperative morbidity, such as postoperative pain, abnormal gait, and hematoma formation, which may be caused by the injury to muscle, fascia, and periosteum during operation.[88] The solutions provided by included studies to the problem are as follows: posterior approach; minimally invasive technique; preemptive analgesia.

Surgical Approaches of Iliac Cancellous Bone Harvesting

Anterior iliac crest is the most common ICBH site, which has been considered the "gold standard."[89] The major benefit of anterior approach is that procedures of donor and recipient site can be done simultaneously, which results in shorter duration of operation and anesthesia time.[90] However, other than the morbidity mentioned above, anterior approach can still result in potential risks of pelvic fracture and sensory nerves injury.[91]

We found insufficient evidence that when compared with anterior approach, posterior approach can reduce the blood loss, length of hospital stay, donor-site morbidity,[45] which might be associated with less muscle reflection require.[92] And the overall cost of posterior approach showed no difference.[46] However, because of the reposition of the patients, operation duration is significantly longer in posterior approach group.[45] It is worth mentioning that bone volume is not the issue that should be considered, since the accessible bone harvested from either anterior or posterior approach is enough for alveolar bone grafting, which has been confirmed by anatomic[93] and clinical[45,92] studies.

Minimally Invasive Technique

Minimally invasive technique is widely applied in bone graft harvesting for its efficiency in alleviating the harm of operations. According to the included studies, minimally invasive technique applied in ICBH mainly includes modified biopsy needle, trephine, bone grinder, and Shepard osteotome. The length of incision is about 0.5 to 2.5 cm, while the incision of the traditional open technique proposed by Wolfe and Kawamoto[94] is usually about 3 and 4 cm.

We found moderate evidence that using the Shepard osteotome, which was first described by Shepard and Dierberg in 1987,[95] could alleviate the morbidity. Moreover, insufficient evidence showed that this method is easy to operate, with no effect on incidence of complications. As for using powered/manual trephine through a 1 to 1.5 cm incision, we found moderate evidence that morbidity is alleviated without extra risk of complications, and the operation is easy to perform. We found insufficient evidence that incidence of morbidity is lower when using the modified biopsy needle through a 5 mm incision or bone grinder through a 1 to 1.5 cm incision. Both of the procedures are easy to perform, and would not cost extra risk of complications.

The common disadvantages of minimally invasive method are the risk of injuring the pelvic organs and extra cost.[47] Since only insufficient evidence was found that using trephine or Shepard osteotomy showed no influence on bone graft survival, it is difficult to determine whether the heat generated by the drill would have impact on the viability of the bone marrow cells. Some clinicians showed certain concern to the volume of cancellous bone harvested through the minimally invasive technique; nevertheless, the minimally invasive technique can finally achieve sufficient volume as much as 20 to 30 cc[53] to satisfy the demand of SABG by harvesting the bone through 1 incision in multiple directions or more incisions when needed.

Preemptive Analgesia

A large portion of patients underwent ICBH experienced significant postoperative pain, which can persist for a long time.[96] Pain can cause adverse physiological effect and emotional distress in both patients and their families.[97] Conventionally, pain following ICBH is well alleviated by the application of patient-controlled analgesia system of opiates. However, opiates may cause side effects such as oversedation, respiratory depression, sleepiness, and nausea.[54] Moreover, both uncontrolled pain and opiates usage can result in longer hospital stay, which may cause more stress, financial cost, and risk of nosocomial infections. To solve these problems, additional analgesics including nonsteroidal anti-inflammatory drugs and local anesthetic agents have been considered.

Ketorolac is a commonly used nonsteroidal anti-inflammatory drug. Its analgesic effect for postoperative pain has been confirmed in both adult and pediatric patients.[98,99] However, we found insufficient evidence that for ICBG postoperative pain, intravenous ketorolac showed no significant effect,[58] which may be caused by limited number of studies included and outcomes reported.

Bupivacaine is applied as local anesthetics for management of donor-site pain after ICBG with variable administrations as follows: infiltration; continuous infusion; block anesthesia; soaked in gelform.

The infiltration of bupivacaine was mainly reported in the 1990s. We found insufficient evidence that it can alleviate the postoperative pain, but it will not last for a long time.[59] The continuous infusion of bupivacaine is delivered through a catheter placed subcutaneously over the ilium crest intraoperatively. We found moderate evidence that bupivacaine infusion could alleviate morbidity. Moreover, insufficient evidence was found that the method would not lead to extra risk of complications. In addition, bupivacaine can be applied in the transversus abdominis block for patients who underwent ICBH on the basis of general anesthesia, and we found moderate evidence that using transversus abdominis plane block can result in milder morbidity, but it will not last for a long time.[56,57]

Dashow et al[60] reported a novel mode of bupivacaine administration by placing a bupivacaine-soaked absorbable gelatin sponge over the ilium crest intraoperatively, which could release bupivacaine continuously. It showed excellent results as significant low postoperative pain score and less morphine usage. But as only 1 study used this method, this evidence was considered to be insufficient. Moreover, it will be considered that it is uncertain whether gelform itself or bupivacaine has effect on pain management, and the pharmacokinetics of gelform is unclear as well.

Ambulatory Alveolar Bone Grafting

To improve the comfort in the recovery stage for patients after SABG and ICBH, the reforms and innovations have been proposed constantly. It has been reported that using preemptive local anesthesia and antiemetic therapy could turn SABG to be an ambulatory surgery,[61] which is beneficial to the psychological status of children patients in the recovery stage. We found insufficient evidence that ambulatory SABG does not lead to extra complications, and the total cost is relevantly low when compared with inpatient SABG. As only 1 study aimed at the complications of ambulatory SABG, more high-quality studies are required for supporting the clinical promotion.

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